

Corrector Timing as Statistical Intervention Diagnosis

A supervisor-facing theory memo

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Abstract

This memo locks the thesis direction around a statistical, rather than algorithmic, theory of corrector timing. The central object is the value of a corrector call as an intervention in an inhomogeneous generative Markov chain. The proposed contribution is a finite-budget diagnostic framework: define corrector-timing estimands, decompose schedule gains into marginal and interaction effects, state safe regret implications when low-order structure is present, and connect the framework to informed correctors, locally balanced MCMC proposals, and non-uniform scan Gibbs. The mathematical claims here are deliberately modest. We do not claim that ProSeCo is locally balanced, that the schedule objective is submodular, or that a search procedure is optimal. The framework is model- and corrector-agnostic; ProSeCo-OWT is one empirical instantiation used to diagnose a non-marginal, redundant, locally exploitable timing regime.

1 Direction lock

The thesis should not be framed as “we designed a better schedule optimizer.” That is too close to a model-specific engineering problem. The more stable statistical question is:

For a fixed masked-diffusion predictor, fixed informed corrector, quality functional F , and finite corrector-call budget B , can we diagnose when corrector timing is marginal, state-dependent, redundant, or interaction-driven?

Algorithms then become probes. A separable ranker probes whether marginal effects are sufficient. Pairwise diagnostics probe whether interactions are structured and composable. Local or true- G search probes whether useful structure remains after low-order summaries fail. Bayesian optimization is a possible downstream method only if the diagnostics reveal a smooth kernel structure; it is not a premise of the thesis.

1.1 The gap

Existing work answers adjacent but different questions.

- Informed-corrector work for discrete diffusion designs better local corrector transitions. Zhao, Shi, Chen, Druckmann, Mackey, and Linderman propose predictor-corrector schemes where the corrector uses model information, including hollow-transformer modifications, to counter approximation error in discrete diffusion sampling [1].
- Locally balanced proposals answer how local MCMC proposals in discrete spaces should use local target information. Zanella characterizes locally balanced proposals and proves Peskun-type optimality in high-dimensional regimes [2].
- Random-scan, non-uniform, and adaptive Gibbs work studies how update schemes or selection probabilities affect MCMC convergence [3, 4].

- Lavenant and Zanella study error bounds and optimal predictor schedule sizes for masked diffusions with factorized approximations [7].

The thesis gap is the timing layer:

Given a fixed predictor and a fixed informed corrector kernel, when should a limited number of corrector calls be deployed along the generation trajectory?

This is not a new corrector-kernel theorem and not a new unmasking-schedule theorem. It is a finite-budget intervention problem over an inhomogeneous sequence of local correction opportunities.

1.2 Candidate theory options

There are three plausible mathematical directions. They are not equally safe.

Option	What it would prove	Assessment
Kernel theorem	ProSeCo or a Zhao-style corrector is locally balanced, contractive, or Peskun-optimal	Too strong for this thesis unless the kernel is redesigned and analysed directly
Optimizer theorem	A particular search or greedy scheduler is optimal or near-optimal	Not aligned with the diagnostic/statistical direction; risks becoming model-specific optimization
Diagnostic theory	Corrector timing has well-defined intervention effects, exact interaction decompositions, and safe low-order implications	Most promising: mathematically clean, corrector-agnostic, and directly connected to the experiments

The rest of this memo develops the third option. The stronger kernel and optimizer theories appear only as intuition or future work.

2 Mathematical setup

Assume a finite or countable state space $\mathcal{X}$. A masked-diffusion sampler with fixed predictor is represented as an inhomogeneous Markov chain

$$X_0 \sim \mu_0, \quad X_t \sim P_t(X_{t-1}, \cdot), \quad t = 1, \dots, T,$$

where P_t is the predictor transition at denoising step t . A fixed corrector at time t is a Markov kernel K_t on $\mathcal{X}$. A schedule is a subset $S \subseteq [T] := \{1, \dots, T\}$. Under schedule S , the law of the terminal sample is

$$\mu_T^S = \mu_0 P_1 K_1^{\mathbf{1}\{1 \in S\}} P_2 K_2^{\mathbf{1}\{2 \in S\}} \dots P_T K_T^{\mathbf{1}\{T \in S\}},$$

where K_t^0 is the identity kernel and $K_t^1 = K_t$. Let $F : \mathcal{X} \rightarrow \mathbb{R}$ be a bounded terminal quality functional. Define

$$J(S) := \mathbb{E}_{\mu_T^S}[F(X_T)], \quad G(S) := J(S) - J(\emptyset).$$

The fixed-budget timing problem considers schedules with $|S| = B$.

Remark 1 (What is abstracted away). *This notation does not require K_t to be Gibbs, Metropolis-within-Gibbs, or locally balanced. It only requires that the corrector call can be represented as a kernel inserted into the trajectory. This is why the framework is corrector-agnostic.*

Definition 1 (Timing estimands). For $t \in [T]$, define the marginal intervention effect

$$\Delta_t := G(\{t\}).$$

For $s \neq t$, define the pair interaction

$$\xi_{s,t} := G(\{s, t\}) - G(\{s\}) - G(\{t\}).$$

For $A \subseteq B \subseteq [T]$ and $x \notin B$, define the diminishing returns gap

$$\text{DR}(A, B, x) := [G(A \cup \{x\}) - G(A)] - [G(B \cup \{x\}) - G(B)].$$

If $\text{DR}(A, B, x) \geq 0$ for all such triples, then G is submodular.

In the experiments, $G_i(S)$ is evaluated on a paired seed i using common random numbers: the base trajectory and scheduled trajectories share all randomness except the corrector interventions. This is a design choice for variance reduction and causal comparability, not an additional theorem assumption.

3 Exact interaction decomposition

The first mathematically clean contribution is to treat G as a finite-set response surface. No approximation is required.

Theorem 1 (Möbius interaction decomposition). Let $G : 2^{[T]} \rightarrow \mathbb{R}$ with $G(\emptyset) = 0$. There exists a unique collection of interaction coefficients $\{I(U) : \emptyset \neq U \subseteq [T]\}$ such that

$$G(S) = \sum_{\emptyset \neq U \subseteq S} I(U) \quad \text{for every } S \subseteq [T].$$

They are given by Möbius inversion:

$$I(U) = \sum_{V \subseteq U} (-1)^{|U|-|V|} G(V).$$

Proof. Define $I(U)$ by the displayed inversion formula for nonempty U , and set $I(\emptyset) = G(\emptyset) = 0$ for this proof. Then

$$\sum_{\emptyset \neq U \subseteq S} I(U) = \sum_{U \subseteq S} \sum_{V \subseteq U} (-1)^{|U|-|V|} G(V) = \sum_{V \subseteq S} G(V) \sum_{W \subseteq S \setminus V} (-1)^{|W|}.$$

The inner sum equals 1 if $V = S$ and 0 otherwise. Hence the expression equals $G(S)$. Uniqueness follows by induction on $|S|$: once all coefficients on strict subsets are known, $I(S) = G(S) - \sum_{U \subsetneq S} I(U)$. $\square$

The first-order coefficients are $I(\{t\}) = \Delta_t$. The second-order coefficients are $I(\{s, t\}) = \xi_{s,t}$. Higher-order coefficients quantify what remains unexplained by marginal and pair effects.

3.1 Diagnostic ladder

The decomposition gives a natural ladder:

Dominant component	Statistical diagnosis	Policy implication
First order $I(\{t\})$	marginal regime	separable ranking can work
Second order $I(\{s, t\})$	redundancy/complementarity	interaction diagnostics needed
Higher order $I(U), U \geq 3$	non-low-order regime	true- G search only as probe

This table is not a theorem about neural networks. It is an exact decomposition plus a diagnostic interpretation.

4 Safe surrogate implications

The next pieces are standard but useful. They say when a diagnostic object would justify a policy class.

4.1 Marginal validity

Define the additive surrogate

$$A(S) := \sum_{t \in S} \Delta_t.$$

Proposition 1 (Marginal regret under uniform additive residual). *Fix B . Suppose*

$$\sup_{|S|=B} |G(S) - A(S)| \leq \eta_B.$$

Let $S_A^ \in \arg \max_{|S|=B} A(S)$ and $S_G^* \in \arg \max_{|S|=B} G(S)$. Then*

$$G(S_G^*) - G(S_A^*) \leq 2\eta_B.$$

Proof. By the residual bound,

$$G(S_G^*) \leq A(S_G^*) + \eta_B \leq A(S_A^*) + \eta_B \leq G(S_A^*) + 2\eta_B.$$

□

If $\hat{\Delta}_t$ is a proxy with $\sup_t |\hat{\Delta}_t - \Delta_t| \leq \varepsilon$, then selecting the top- B times by $\hat{\Delta}_t$ adds at most $2B\varepsilon$ regret on A , giving the familiar $2\eta_B + 2B\varepsilon$ bound. This is the right theorem for marginal rankers. If the bound is vacuous or the empirical rankers fail, the regime is not marginal.

4.2 Pairwise validity

Define the second-order surrogate

$$Q(S) := \sum_{t \in S} \Delta_t + \sum_{\{s,t\} \subseteq S} \xi_{s,t}.$$

Proposition 2 (Pairwise surrogate regret). *Suppose*

$$\sup_{|S|=B} |G(S) - Q(S)| \leq \zeta_B.$$

Let $S_Q^ \in \arg \max_{|S|=B} Q(S)$. Then*

$$G(S_G^*) - G(S_Q^*) \leq 2\zeta_B.$$

If an estimated surrogate $\hat{Q}$ satisfies $\sup_{|S|=B} |Q(S) - \hat{Q}(S)| \leq \alpha_B$, and $\hat{S}$ maximizes $\hat{Q}$, then

$$G(S_G^*) - G(\hat{S}) \leq 2\zeta_B + 2\alpha_B.$$

Proof. The first statement is identical to the marginal proof with Q replacing A . For the estimated case,

$$Q(\hat{S}) \geq \hat{Q}(\hat{S}) - \alpha_B \geq \hat{Q}(S_Q^*) - \alpha_B \geq Q(S_Q^*) - 2\alpha_B.$$

Thus $\hat{S}$ has Q -optimization gap at most $2\alpha_B$. Apply the exact- Q argument with this optimization gap. □

This is not a deep optimization theorem. Its value is diagnostic: it states precisely what must be true before a pairwise theory supports scheduling.

5 Redundancy and diminishing returns

The pairwise model gives a clean connection between negative interactions and diminishing returns.

Proposition 3 (Second-order submodularity criterion). *Let*

$$Q(S) = \sum_{t \in S} a_t + \sum_{\{s,t\} \subseteq S} b_{s,t}.$$

Then Q is submodular if and only if $b_{s,t} \leq 0$ for every pair $\{s, t\}$.

Proof. For $A \subseteq B$ and $x \notin B$,

$$Q(A \cup \{x\}) - Q(A) = a_x + \sum_{y \in A} b_{x,y},$$

and similarly for B . Therefore

$$\text{DR}_Q(A, B, x) = - \sum_{y \in B \setminus A} b_{x,y}.$$

This is nonnegative for all A, B, x if all $b_{x,y} \leq 0$. Conversely, take $A = \emptyset$, $B = \{y\}$, $x \neq y$. Then $\text{DR}_Q(\emptyset, \{y\}, x) = -b_{x,y}$. Submodularity implies $b_{x,y} \leq 0$. $\square$

Corollary 1 (Robustness to higher-order residuals). *Write $G = Q + R$. For any $A \subseteq B$, $x \notin B$,*

$$\text{DR}_G(A, B, x) = \text{DR}_Q(A, B, x) + \text{DR}_R(A, B, x).$$

If $\sup_S |R(S)| \leq \rho$, then

$$|\text{DR}_R(A, B, x)| \leq 4\rho.$$

Hence a positive second-order diminishing-returns gap larger than 4ρ survives the higher-order residual.

Proof. The first identity follows by linearity. The second follows because DR_R is a signed sum of four R -values. $\square$

This corollary is useful because it tells us why negative pair interactions alone do not prove submodularity of G . They prove submodularity of the second-order approximation. To transfer the conclusion to G , the higher-order residual must be controlled.

6 Corrector value as error, correctability, and downstream relevance

The previous sections are exact finite-set/statistical statements. They do not explain why a corrector call is valuable. For that, we introduce an idealized Markov-chain layer.

At time t , suppose there is an ideal local target π_t and current law ν_t . The corrector kernel K_t is useful if it contracts discrepancy from π_t . Assume an entropy contraction condition:

$$\text{KL}(\nu K_t \parallel \pi_t) \leq (1 - \kappa_t) \text{KL}(\nu \parallel \pi_t), \quad \kappa_t \in [0, 1].$$

This assumption is inspired by Markov-chain convergence theory, including entropy contraction for Gibbs samplers under log-concavity [5], but it is *not* asserted for ProSeCo without separate verification.

Proposition 4 (Clean contraction-index statement). *If the local score of interest is negative relative entropy*

$$\Phi_t(\nu) := -\text{KL}(\nu \parallel \pi_t),$$

and K_t satisfies the contraction condition above, then applying the corrector improves this idealized score by at least

$$\Phi_t(\nu K_t) - \Phi_t(\nu) \geq \kappa_t \text{KL}(\nu \parallel \pi_t).$$

Proof.

$$\Phi_t(\nu K_t) - \Phi_t(\nu) = \text{KL}(\nu \parallel \pi_t) - \text{KL}(\nu K_t \parallel \pi_t) \geq \kappa_t \text{KL}(\nu \parallel \pi_t).$$

□

This is the clean form of the intuition:

$$\text{ideal corrector value} \gtrsim \text{residual error} \times \text{local contraction strength}.$$

For a terminal quality functional F , one also needs downstream relevance. Let $h_t(x)$ be the expected terminal reward from state x at time t under the remaining predictor/corrector plan. If h_t is bounded, then Pinsker’s inequality gives

$$|(\nu K_t - \pi_t)h_t| \leq \text{osc}(h_t) \sqrt{\frac{1}{2} \text{KL}(\nu K_t \parallel \pi_t)} \leq \text{osc}(h_t) \sqrt{\frac{1 - \kappa_t}{2} \text{KL}(\nu \parallel \pi_t)}.$$

This does not prove the sign of the terminal improvement. It says that a contractive corrector reduces a bound on downstream bias relative to the ideal local target, with strength governed by three factors:

$$\boxed{\text{residual error } \text{KL}(\nu_t \parallel \pi_t) \quad \times \text{correctability } \kappa_t \quad \times \text{downstream sensitivity } \text{osc}(h_t)}.$$

This explains why uncertainty signals alone can fail. High entropy can reflect missing context rather than correctable error; a large residual is not useful if K_t cannot contract it; and even a successful local correction has little terminal value if downstream sensitivity is low.

7 Connection to locally balanced proposals

Zanella’s locally balanced proposals are a theory for constructing informed local MCMC moves in discrete spaces. In a neighborhood $\mathcal{N}(x)$ of state x , a locally informed proposal has the form

$$q_g(y \mid x) \propto g\left(\frac{\pi(y)}{\pi(x)}\right) K_0(x, y), \quad y \in \mathcal{N}(x),$$

where K_0 is a base local proposal and g satisfies a balancing condition of the form $g(r) = r g(1/r)$. Zanella shows that locally balanced choices are optimal in a precise high-dimensional Peskun sense [2].

The thesis should use this in the following disciplined way.

- Locally balanced proposals answer the *kernel-design* question: given a local target, how should the sampler propose local moves?
- Informed correctors for discrete diffusion answer a related generative-model question: how can a corrector use learned model information to counter predictor approximation error?
- Corrector timing asks a different question: given such a local kernel K_t , when should a finite number of calls to K_t be spent along an inhomogeneous trajectory?

Thus the clean connection is:

An informed corrector is a locally informed Markov kernel. Corrector timing is the finite-budget scan problem of deploying such kernels along a generative trajectory.

We should not claim that ProSeCo is locally balanced unless this is separately proved. The theory only requires that K_t be a well-defined correction kernel and that its statistical effect can be estimated.

8 Connection to non-uniform scan Gibbs

Random-scan Gibbs chooses which coordinate or block to update. The selection probabilities affect convergence. Roberts and Sahu analyze how updating schemes, blocking, and covariance structure affect Gibbs behavior [3]. Chimisov, Latuszynski, and Roberts develop an adaptive random-scan Gibbs sampler that adapts selection probabilities using a spectral-gap criterion for a Gaussian analogue [4].

The corrector-timing analogy is:

Gibbs / MCMC concept	Corrector-timing analogue
coordinate or block	denoising time / correction opportunity
local update kernel	corrector kernel K_t
scan probability	timing policy or budget allocation
mixing improvement	terminal gain $G(S)$ or local contraction index
adaptive scan	state-dependent corrector timing

The analogy is useful but not literal. A Gibbs sampler targets one stationary distribution. A masked-diffusion sampler is an inhomogeneous generative chain whose intermediate targets change with time. Therefore scan Gibbs supplies vocabulary and motivation, not an off-the-shelf theorem.

In the additive idealization, if a single call at time t has value c_t , then allocating B deterministic calls to the largest c_t 's is optimal. The statistical content of this thesis is precisely that real corrector calls need not remain additive: early calls change later states, later values can become redundant, and pair/higher-order effects determine the regime.

9 Research question and theorem stack

The recommended locked research question is:

For a fixed masked-diffusion predictor and fixed informed corrector, can finite-budget corrector timing be statistically diagnosed from marginal effects, conditional trajectory state, and interaction effects; and when is timing reducible to marginal ranking versus interaction-driven correction?

The corresponding theory stack is:

1. **Intervention setup.** Define $J(S)$, $G(S)$, Δ_t , $\tau_t(z)$, $\xi_{s,t}$, and $\text{DR}(A, B, x)$.
2. **Exact interaction decomposition.** Use Möbius inversion to define marginal, pairwise, and higher-order effects exactly.
3. **Safe low-order implications.** Prove marginal and pairwise surrogate regret bounds under explicit finite-pool residual assumptions.

4. **Redundancy/submodularity diagnostic.** Show negative pair coefficients imply submodularity of the second-order model, and quantify the higher-order residual needed to transfer this conclusion to G .
5. **Correctability index.** Under an idealized contraction assumption, derive that local corrector value is governed by residual error, contraction strength, and downstream sensitivity.
6. **Regime classification.** Classify a model/corrector/ F/B tuple as no-op, marginal, redundant/pairwise, locally searchable, or higher-order/black-box.

10 How ProSeCo-OWT fits

ProSeCo-OWT should be used as one empirical case study, not as universal evidence. The current diagnostics say:

- Corrector timing has positive headroom over uniform timing.
- Tested separable rankers fail, so the regime is not purely marginal.
- Pair interactions are structured and mostly negative, indicating redundancy.
- Cheap pairwise surrogates help at $B = 2$ but fail at $B = 3, 4$, so low-dimensional pairwise composition is insufficient.
- Exact diminishing-return gaps are mildly positive on average, but not strong enough for a clean submodularity claim.
- One-swap neighborhoods contain many improvements, so local true- G search is a useful probe.
- Bayesian optimization remains unclear because the schedule-distance kernel signal is weak.

Therefore ProSeCo-OWT is best described as an *interaction-driven, locally exploitable corrector-timing regime*. This is an empirical classification inside a model-agnostic framework.

11 What is proved, what is borrowed, what is intuition

Item	Status in this thesis	Source / inspiration
Intervention $G, \Delta, \xi, \text{DR}$	Defined here; mathematically direct	potential-outcome/statistical response-surface view-point
Möbius decomposition	Exact theorem	finite Boolean-lattice inversion
Marginal and pairwise regret bounds	Exact under explicit sup residual assumptions	standard surrogate-regret argument
Second-order submodularity criterion	Exact for quadratic set functions	elementary set-function algebra
Contraction-index proposition	Exact under stated KL contraction assumption	Markov-chain contraction/Gibbs theory intuition
Locally balanced proposal connection	Conceptual bridge; not assumed for ProSeCo	Zanella informed proposals
Scan Gibbs connection	Conceptual bridge; not theorem transfer	Roberts-Sahu; adaptive scan Gibbs
ProSeCo-OWT classification	Empirical case study only	current diagnostics

12 Recommended thesis claim

A careful final claim would be:

This thesis develops a statistical diagnostic framework for finite-budget corrector timing in masked diffusion. It treats corrector calls as interventions in an inhomogeneous generative Markov chain, decomposes schedule gain into marginal and interaction effects, and connects corrector timing to the scan-design problem for locally informed Markov kernels. On ProSeCo-OWT, the framework identifies a non-marginal, redundant, locally exploitable regime: separable marginal timing fails, pair interactions are mostly negative, weak diminishing returns are present but not theorem-grade, and true- G search is useful as an empirical probe.

Claims to avoid:

- ProSeCo is locally balanced.
- Corrector timing is submodular in general.
- Bayesian optimization is justified by current diagnostics.
- ProSeCo-OWT proves model-agnostic empirical behavior.
- Search is the theoretical contribution.

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